Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
3rd Year 2nd Semester Final Examination, 2024
Course Code: EEE 337, Session: 2020-2021
Course Title: Power System II

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Answer any two questions from each part

Part A

1. (a) Define Inductance with the required expression. 1
   (b) Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. 4
   (c) Derive the expression of Capacitance of a three phase line with unsymmetrical spacing. 10
   (d) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. 10

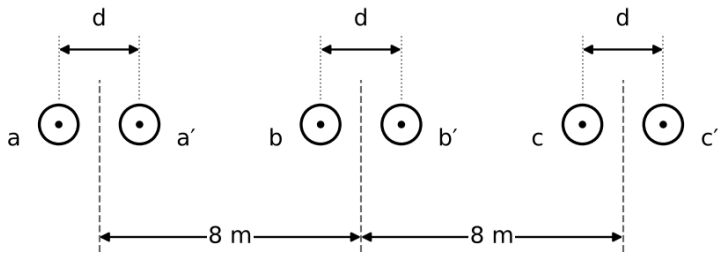


Fig. 1 of the paper (redrawn): figure for Q-1(b)

2. (a) Define Corona Effect. 1
   (b) Define Transposed Transmission line. Explain the reason behind Transpose. 4
   (c) Derive the expression of Inductance of composite conductor lines. 10
   (d) A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. 10

3. (a) Define Skin Effect. 1
   (b) Describe different FACTS devices and applications. 4
   (c) Derive the expression of dielectric stress in a single core cable. 10
   (d) The capacitances of a 3-phase belted cable are $12 \cdot 6$ μF between the three cores bunched together and the lead sheath and $7 \cdot 4$ μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. 10

Part B

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4. (a) Define critical disruptive voltage. Write down its significance. 5

   (b) What is String Efficiency? Discuss the methods of improving String Efficiency. 10

   (c) The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage 10

5. (a) What are the causes of insulation failure? 5

   (b) Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end. 10

   (c) A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. 10

6. (a) Define Power Angle. Briefly explain Equal Area Criterion. 5

   (b) Explain what happens for sudden increase in mechanical input to generator using Pe-δ diagram. 10

   (c) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V|=1.0 pu), with the generator terminal voltage (|Vt|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. 10

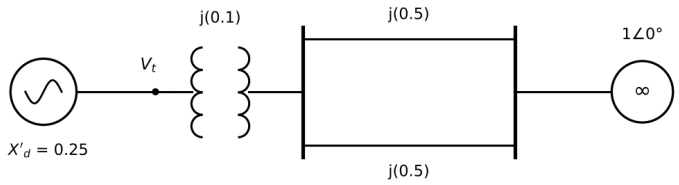


Fig. 2 of the paper (redrawn; the same figure as in the 2021-22 paper): figure for Q-6(c)

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